



Measuring with Sine Bar

Gauges are used to find the angle in this method firstly sine bar this placed on the surface plate. Now, the sine bar is placed on the taper side of the object. Then slip gauges are arranged one by one upwards till touch the right side of the roller. The remove the object from the sine bar. Now find the total height of slip gauge and length of sine bar (size of sine bar)

$$Sin\theta = \frac{Opposite}{Hypotenuse} = \frac{h}{l}$$

where, h – Total height of slip gauges, l – Length of sine bar

3.8 GAUGES

Gauges are the instruments which are ready made with required size and shape .We use this gauge instantly to find out the Size and shape. There is no graduated part in the gauges. They are made up of alloy steel and heat treated and fine finishing done by grinding process.

Advantage of Gauges

1. The measurements are checked quickly and easily.
2. The cost of gauges is less when compared with precision measuring instruments.
3. A semi skilled operator easily can handle gauges.
4. No Supervision is required.
5. The production is increased.

GRADES OF GAUGES

Gauges are made up of alloy steel they are hardened and tempered and made in following three different grades

1. Workshop Gauge
2. Inspection Gauge
3. Master Gauge

Workshop Gauge: This grade of gauges is used in workshops to check the products manufacture in the shop with low accuracy, the accuracy of workshop gauge is 0.01 inch or 0.025 mm.

Inspection Gauge: Inspection Ganges are deigned to be handled by skilled operators for inspection purpose. It is made with an accuracy of 0.001 inch or 0.0025 mm.

Master Gauge: Master gauge are useful for checking the workshop gauges and inspection gauges. It is also used to check very accurate tools and is made with an accuracy of 0.00001 inch.

GM_Chp03.indd 46 14-02-2019 11:33:20